CHAPTER 9 · REFERENCE DEPTH

Radio Wave Propagation

A transmitter radiates energy; a receiver picks it up — but how the wave gets from one to the other decides everything about the radio. Will it hug the curve of the Earth, bounce off a layer 300 km up, or shoot in a straight line and stop dead at the horizon? The answer depends almost entirely on frequency.

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SYLLABUS MAP

Part II (ii) Propagation · day/night frequencies · skip distance · fading · ground shadow

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◎ Learning objectives — by the end of this chapter you will be able to...

- Describe the three propagation paths — ground, sky and space wave — their mechanisms and bands.
- Explain the structure of the ionosphere and its daily, seasonal and solar variation.
- Define MUF, LUF and the optimum working frequency, and explain day/night frequency choice.
- Explain skip distance and dead space, with the geometry.
- Calculate line-of-sight range from the radio-horizon formula.

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- Explain attenuation, fading and ground shadow.

9.1 The three paths

9.2 The ground wave

9.3 The ionosphere

9.4 The sky wave: MUF, LUF & OWF

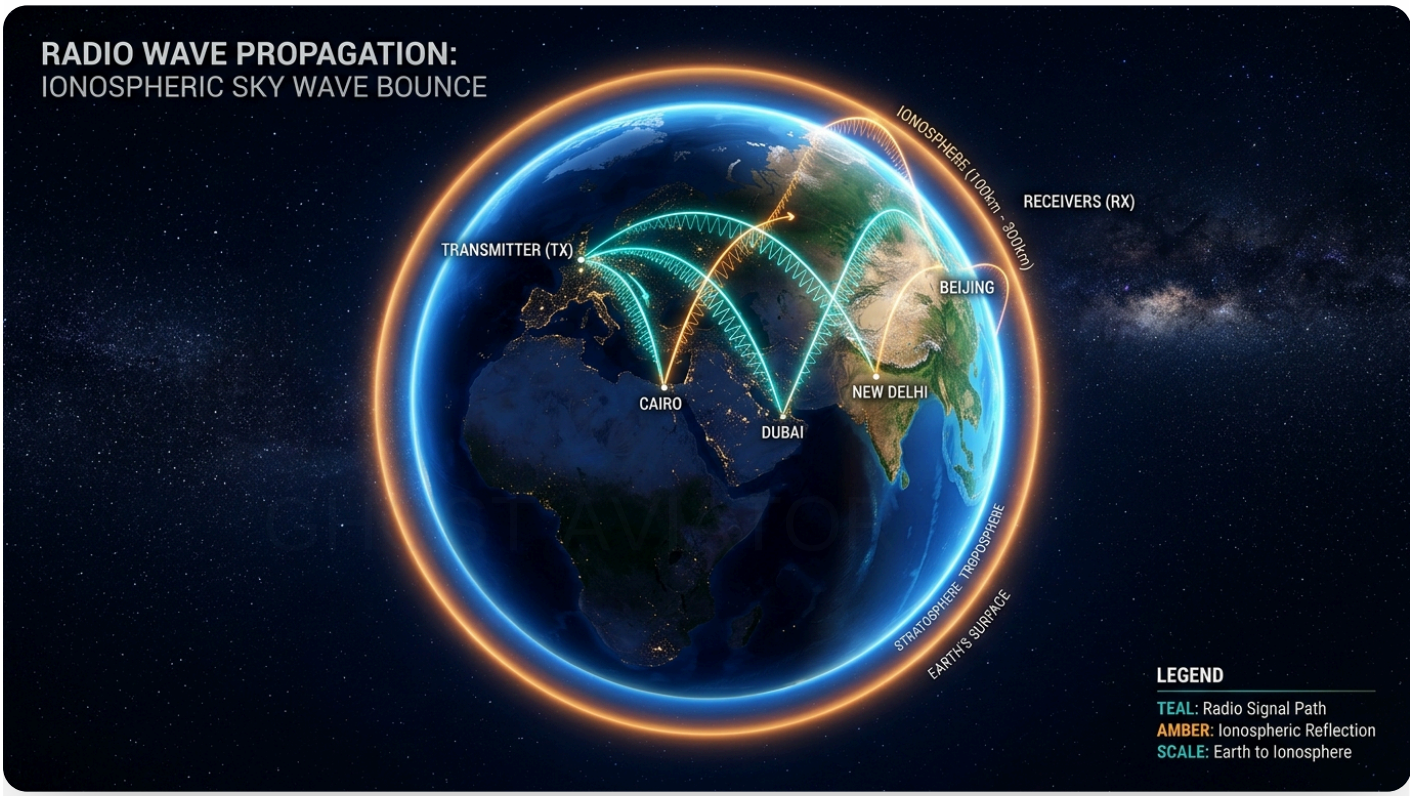
9.5 Day & night frequencies

9.6 Skip distance & dead space

9.7 The space wave & radio horizon

9.8 Attenuation, fading & ground shadow

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The path a radio wave takes across the globe is dictated entirely by its frequency and the physical environment.

9.1 The three paths a radio wave takes

Propagation Paths			
Path	**How it travels**	**Band**	**Used by**
Ground (surface) wave	Follows the curve of the Earth, hugging the surface	LF / MF	NDB; maritime
Sky wave	Refracted (bent back) by the ionosphere	HF	Long-range / oceanic HF voice
Space wave	Straight line (line-of-sight) to the radio horizon	VHF & above	ATC VHF voice, VOR, ILS

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MNEMONIC

Low hugs, High bounces, VHF shoots straight. LF/MF follow the ground; HF bounces off the ionosphere; VHF and up go line-of-sight.

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9.2 The ground wave

FIRST PRINCIPLES — WHY LOW FREQUENCIES FOLLOW THE EARTH

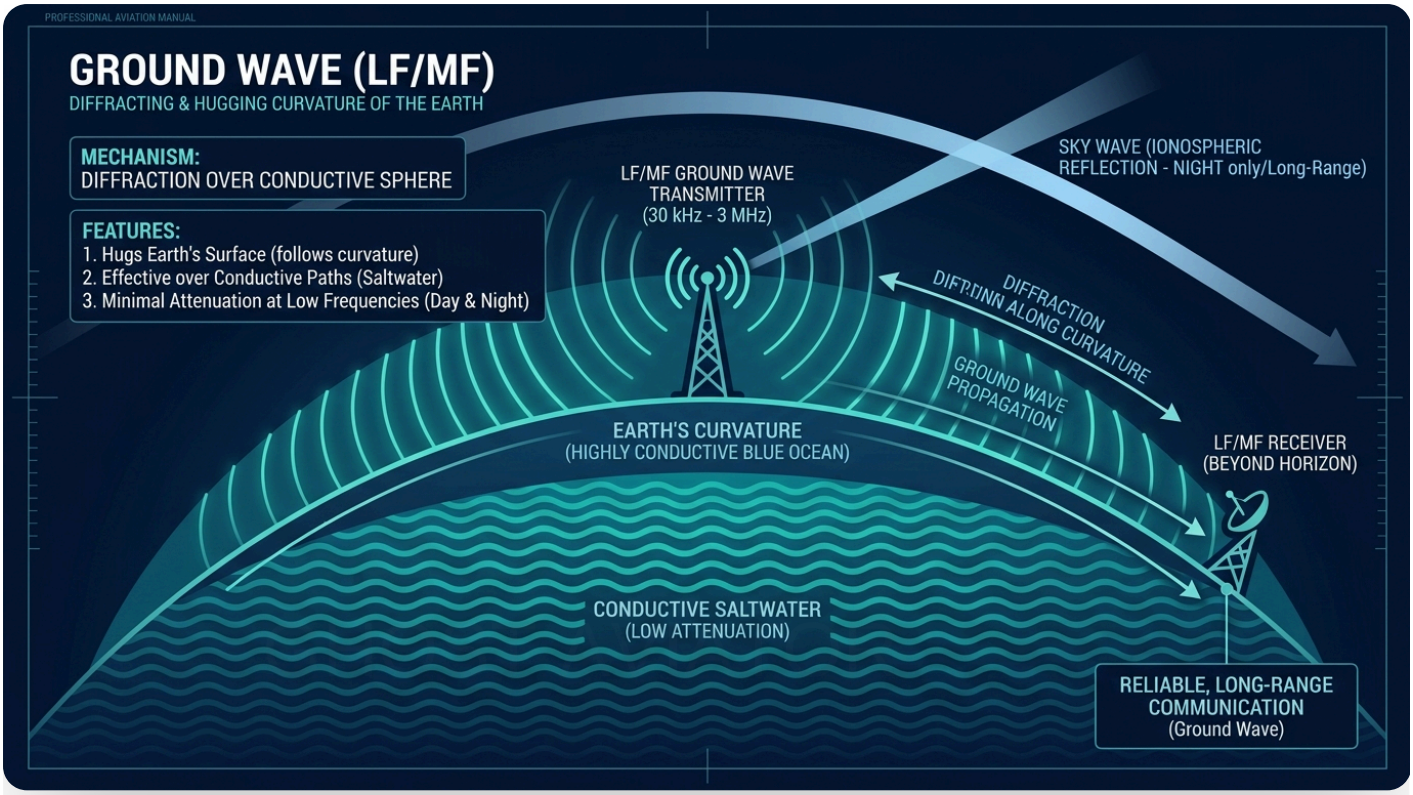
A vertically-polarized wave at low frequency induces currents in the conducting surface of the Earth; these slow the bottom of the wavefront, tilting it forward so it **diffracts around the curve** of the planet. The lower the frequency, the better it clings — and the farther it reaches.

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WHAT SETS GROUND-WAVE RANGE

Range grows with **transmitter power**, with **lower frequency**, and with the **conductivity of the surface** — so it is much greater over **sea water** (highly conductive) than over dry land or mountains. This is the path the NDB relies on, giving useful range beyond line-of-sight at low level.

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The ground wave diffracts around the curvature of the Earth, achieving greatest range over highly conductive sea water.

9.3 The ionosphere

IN PLAIN TERMS

High above the Earth, solar radiation strips electrons from thin air, creating electrically charged **layers**. These act like a mirror for HF radio, bending the wave back to Earth and giving HF its enormous range.

THE LAYERS

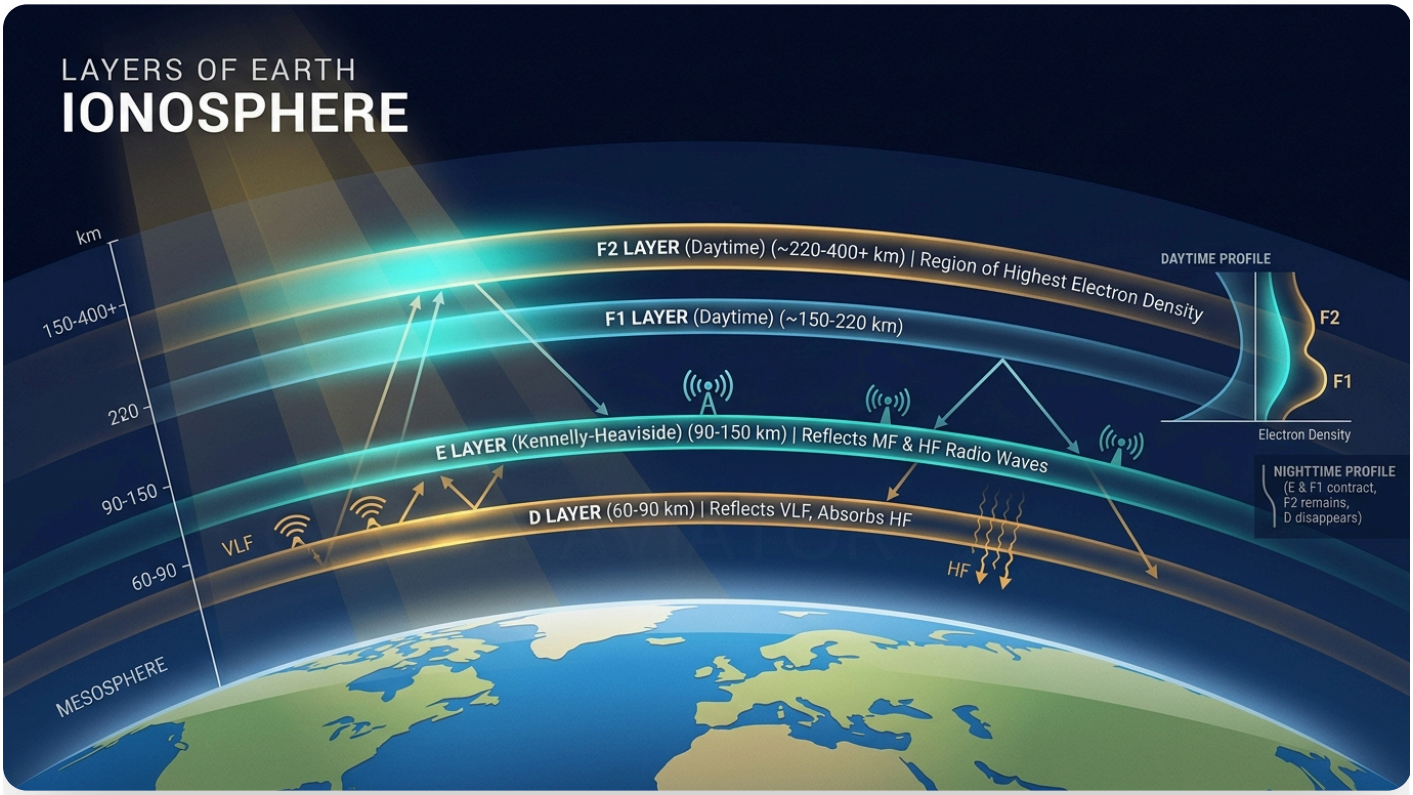
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D layer (lowest, ≈60–90 km) — forms only by **day** and **absorbs** lower HF frequencies. **E layer** (≈90–150 km). **F1 & F2 layers** (highest, ≈150–400 km) — the main HF-refracting layers; at night F1 and F2 merge into a single, weaker F layer.

⚙ VARIATION — WHY HF IS NEVER CONSTANT

Ionisation depends on the Sun, so it varies **diurnally** (strong by day, weak by night), **seasonally**, and with the **11-year solar cycle** (sunspots). The **critical frequency** is the highest frequency a layer will reflect at vertical incidence; above it the wave punches straight through.

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The structure of the ionosphere: The D layer absorbs signals during the day, while the F layers act as a mirror for high-frequency (HF) sky waves.

9.4 The sky wave: MUF, LUF & OWF

DEFINITIONS

MUF — Maximum Usable Frequency: the highest frequency the ionosphere will still refract back to a given point. Above it, the wave escapes into space and is lost.
LUF — Lowest Usable Frequency: the lowest frequency that still arrives usable; below it, D-layer absorption kills it.
OWF — Optimum Working Frequency: typically about **85% of the MUF** — set safely below the MUF so normal ionospheric variation doesn't push the working frequency above it. The usable window lies **between LUF and MUF**.

ANGLE OF INCIDENCE & HOPS

A wave striking the ionosphere at a **shallow (grazing) angle** is refracted back over a long distance; a steep angle may punch through. By bouncing between the ionosphere and the ground, a sky wave can make several **hops**, reaching right around the world.

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9.5 Day & night frequencies

FIRST PRINCIPLES — WHY HF TRACKS THE SUN

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By **day** the D layer is present and absorbs lower frequencies, so you must use a **higher** HF frequency to get through. By **night** the D layer disappears and the ionosphere weakens, so a **lower** HF frequency works best.

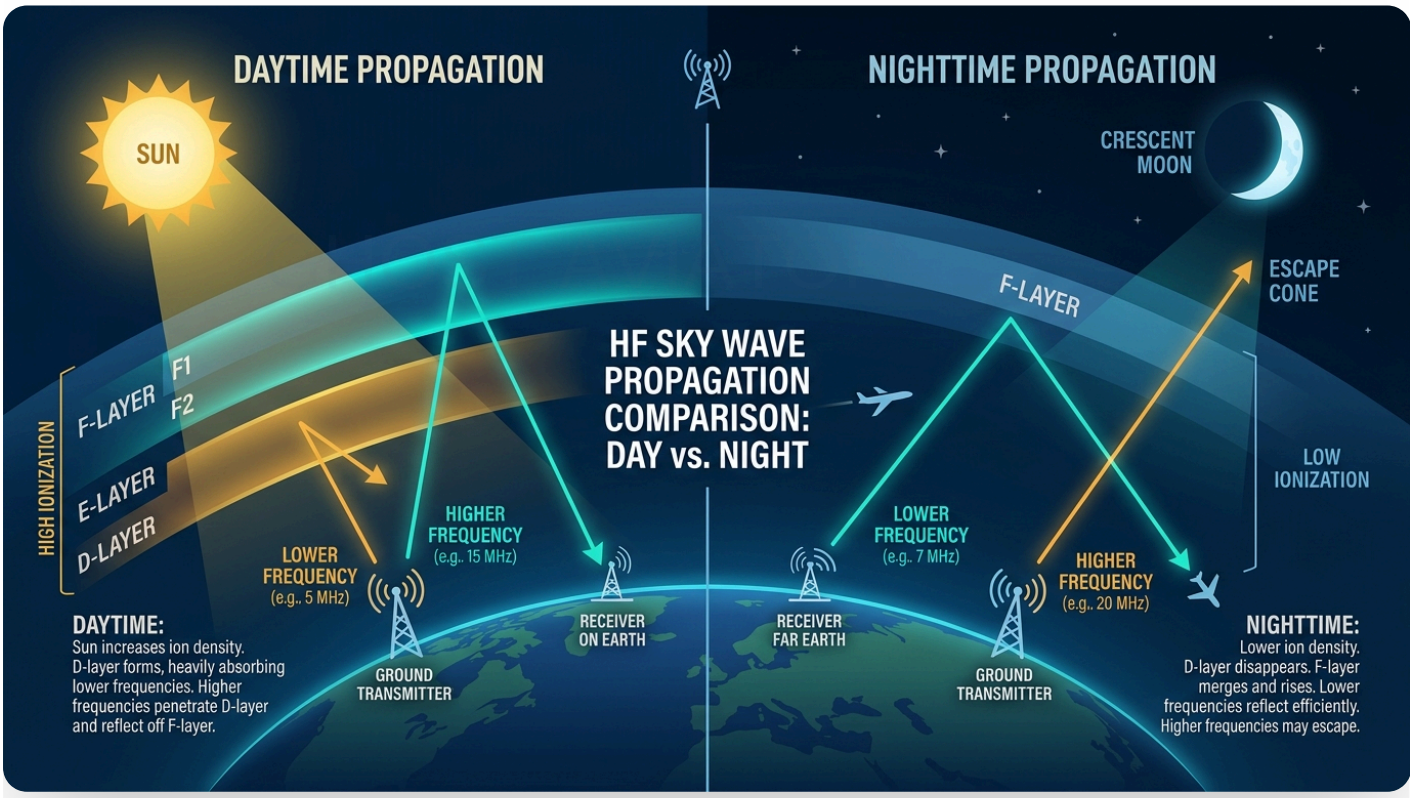
THE RULE OF THUMB

Higher frequency by DAY, lower frequency by NIGHT for HF sky-wave communication. This is why long-haul crews change their HF frequency as they cross the day/night terminator.

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MNEMONIC

"Day = High, Night = Low." Sun up, frequency up.



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By day, the absorbing D-layer forces crews to use higher frequencies. By night, the D-layer vanishes, allowing lower frequencies to reach the reflecting F-layer.

9.6 Skip distance & dead space

DEFINITIONS

Skip distance: the distance from the transmitter to the point where the **first sky wave** returns to Earth.

Dead space (skip zone): the silent ring *between* the end of ground-wave coverage and the point where the first sky wave lands — a region where neither wave reaches and reception fails.

EXAM TRAP

Skip distance **increases** as you raise the frequency (toward the MUF) and as the ionosphere weakens at night — which is why dead space changes through the day. Above the MUF there is no sky-wave return at all.

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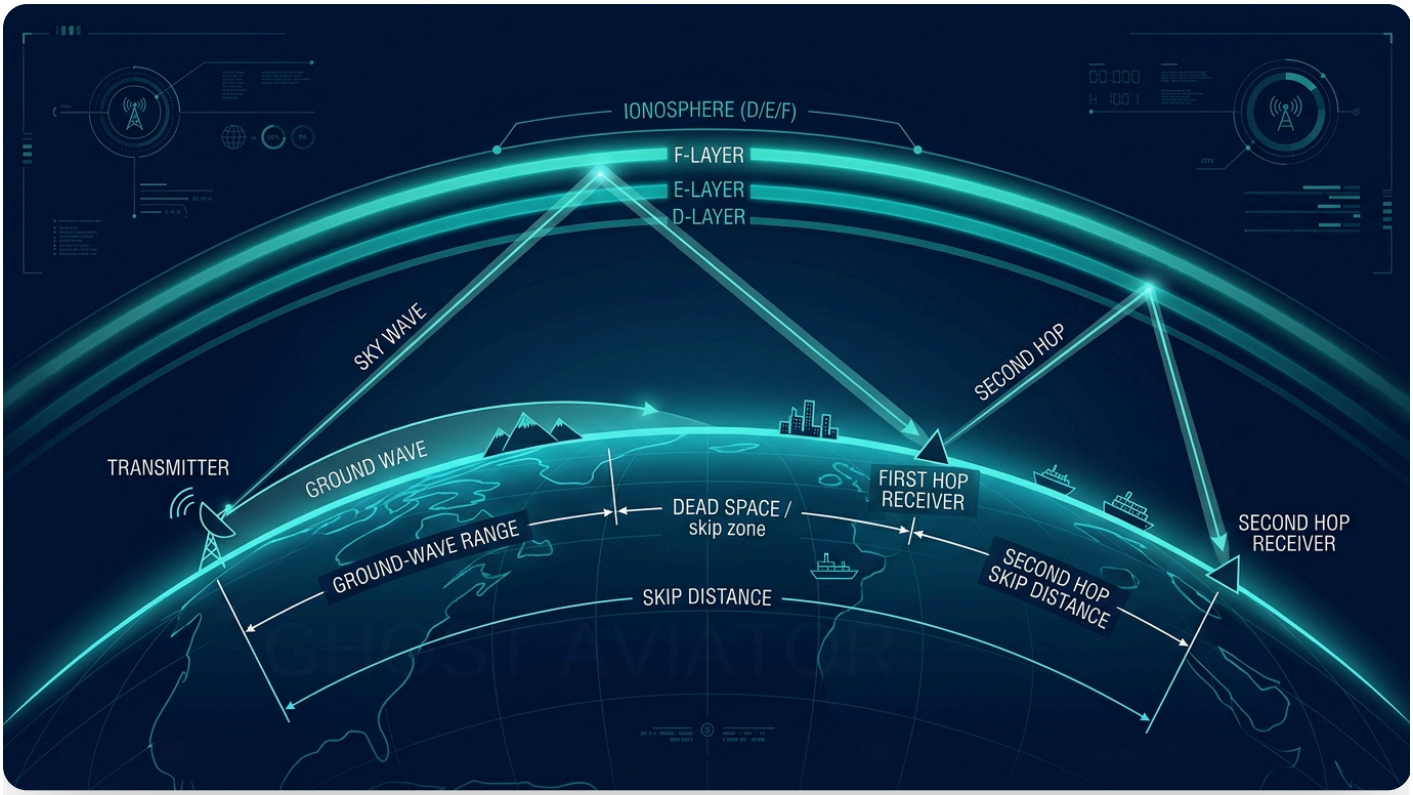


Figure 9.1 — Ground wave, sky wave, skip distance & dead space.

9.7 The space wave & radio horizon

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IN PLAIN TERMS

VHF and above travel in essentially straight lines (the **space wave**), so range is set by the **radio horizon** — and the higher you (or the station) are, the farther that horizon. This is why a VHF call that fails on the ground works once you climb.

$$d\ (\text{NM}) \approx 1.23 \times \sqrt{h}$$

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h = height in feet · the radio horizon is slightly beyond the visual horizon (atmospheric refraction)

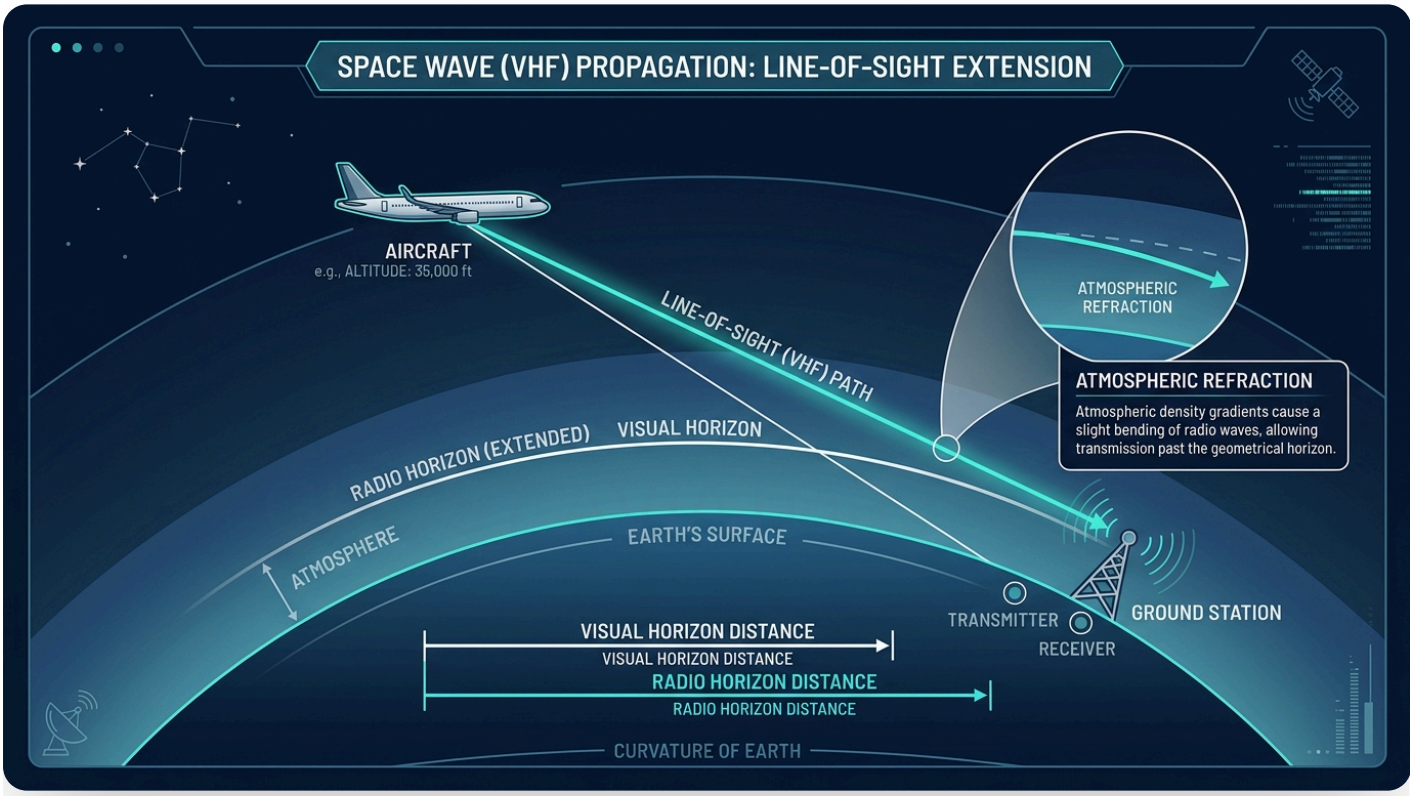
WORKED EXAMPLE — LINE-OF-SIGHT RANGE

At **10,000 ft:** d ≈ 1.23 × √10,000 = 1.23 × 100 = **123 NM** to the radio horizon.

If the ground station antenna is at **400 ft:** its horizon ≈ 1.23 × √400 = 1.23 × 20 = ≈ **25 NM**. The **total** path is the sum: 123 + 25 ≈ **148 NM**.

Takeaway: **more height, more range** — the exact figure isn't usually required, but the principle and the formula are.

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VHF and UHF waves travel in a straight line-of-sight. Atmospheric refraction bends them slightly, making the radio horizon ≈15% farther than the visual horizon.

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▼ Deep dive — ducting & super-refraction

Occasionally a temperature inversion traps VHF/UHF in a "duct" near the surface, carrying it far beyond the normal horizon (super-refraction). The opposite, sub-refraction, shortens range. These are anomalies, not the rule — for the exam, treat VHF as reliable line-of-sight whose range grows with altitude.

9.8 Attenuation, fading & ground shadow

THREE SIGNAL ENEMIES

Attenuation — the natural weakening of a signal with distance and absorption.
Fading — fluctuation in received strength, typically when two paths (ground + sky, or two sky paths) arrive together and interfere; common on HF at dusk.
Ground shadow — loss of a line-of-sight (space) wave behind a hill, building or terrain that blocks it.

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COCKPIT REALITY

Low on approach behind high ground, your VHF can drop out — that's ground shadow; climb a little and the line-of-sight clears. On HF over the ocean at night you'll hear the signal swell and fade — that's sky-wave fading, eased by selecting a better frequency.

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☆ Numbers to memorise

Essential Facts for Chapter 9	
Fact	**Value**
Ground wave	LF/MF, follows Earth's curve, better over sea (NDB)
Sky wave	HF, refracted by ionosphere (long range, multi-hop)
Space wave	VHF+, line-of-sight to radio horizon
Ionosphere layers	D, E, F1, F2 (D absorbs by day; F1/F2 merge at night)
HF day/night	Higher freq by day, lower by night
MUF / LUF / OWF	Max usable / Lowest usable / ≈85% of MUF
Dead space	Gap between ground-wave range and first sky-wave return
Radio horizon	d (NM) ≈ 1.23 × √h(ft)

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Question bank

Part A — MCQs (click an option to check)

1. An NDB relies mainly on which propagation path?

Ground (surface) wave

Sky wave

Space wave

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Satellite

Answer: Ground (surface) wave.

NDB (LF/MF) uses the ground wave that follows the Earth's curvature.

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2. Ground-wave range is greatest over:

Dry desert

Sea water (high conductivity)

Mountains

Forest

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Answer: Sea water (high conductivity).

Higher surface conductivity (sea) gives much greater ground-wave range.

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3. Long-range HF communication is achieved by:

Ground wave

Sky wave refracted by the ionosphere

Line-of-sight space wave

Direct wave only

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Answer: Sky wave refracted by the ionosphere.

HF sky waves refract off the ionosphere to travel thousands of miles.

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4. Which ionospheric layer absorbs lower HF frequencies by day?

D layer

E layer

F1 layer

F2 layer

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Answer: D layer.

The D layer forms by day and absorbs lower frequencies; it disappears at night.

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5. For HF, the best night-time frequency compared with daytime is generally:

Higher

Lower

The same

Zero

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Answer: Lower.

The D layer disappears at night, so a lower frequency works best — "day high, night low."

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6. A frequency above the MUF will:

Be absorbed by the D layer

Pass through the ionosphere into space

Follow the ground

Always reflect

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Answer: Pass through the ionosphere into space.

Above the Maximum Usable Frequency the wave is no longer refracted back — it escapes to space.

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7. The optimum working frequency (OWF) is typically about:

50% of the MUF

85% of the MUF

110% of the MUF

Equal to the LUF

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Answer: 85% of the MUF.

The OWF is set around 85% of the MUF so variation doesn't push it above the MUF.

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8. The silent ring between ground-wave coverage and the first returning sky wave is the:

Skip distance

Dead space (skip zone)

Radio horizon

Fading zone

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Answer: Dead space (skip zone).

Dead space is the gap where neither ground nor sky wave is received.

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9. Skip distance generally increases when:

Frequency is lowered

Frequency is raised toward the MUF / at night

Power is reduced

The antenna is shortened

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Answer: Frequency is raised toward the MUF / at night.

Higher frequency and a weaker (night) ionosphere lengthen the skip distance.

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10. The line-of-sight range from 10,000 ft is approximately:

12 NM

123 NM

1230 NM

40 NM

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Answer: 123 NM.

d ≈ 1.23 × √10,000 = 1.23 × 100 = 123 NM.

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11. VHF reception is lost behind a hill on approach. This is:

Sky-wave fading

Skip distance

Ground shadow

Absorption by the D layer

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Answer: Ground shadow.

Terrain blocking the line-of-sight (space) wave creates a ground shadow.

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12. Fading on an HF signal is usually caused by:

High transmitter power

Two paths interfering (e.g. ground + sky, or two sky paths)

A short antenna

Vertical polarization

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Answer: Two paths interfering (e.g. ground + sky, or two sky paths).

Fading is interference between signals arriving by different paths, common on HF.

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13. The ionosphere's ionisation varies with:

Transmitter power only

Time of day, season and the 11-year solar cycle

Antenna length

Aircraft altitude

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Answer: Time of day, season and the 11-year solar cycle.

It is driven by the Sun, so it varies diurnally, seasonally and with the sunspot cycle.

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14. A sky wave can reach right around the world by:

Following the ground

Multiple hops between the ionosphere and the ground

Travelling in a straight line

Using a satellite

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Answer: Multiple hops between the ionosphere and the ground.

Repeated refraction and ground reflection make multiple hops possible.

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15. The radio horizon is, compared with the visual horizon:

Shorter

Slightly farther (atmospheric refraction)

Exactly the same

Half as far

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Answer: Slightly farther (atmospheric refraction).

Atmospheric refraction bends the wave slightly, extending the radio horizon a little beyond the visual one.

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16. Which path does VHF ATC voice use?

Ground wave

Sky wave

Space wave (line-of-sight)

All three equally

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Answer: Space wave (line-of-sight).

VHF uses the space wave — line-of-sight, range growing with altitude.

Part B — Oral / viva (tap to reveal model answers)

▼ ? **Name the three propagation paths and the bands that use them.**

Model Answer:

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Ground (surface) wave — LF/MF, follows the Earth's curve; sky wave — HF, refracted by the ionosphere; space wave — VHF and above, line-of-sight.

▼ ? **Describe the ionosphere and how it varies.**

Model Answer:

Layers of ionised air — D, E, F1, F2 — created by solar radiation. The D layer forms by day

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and absorbs lower HF; the F layers refract HF. Ionisation varies diurnally, seasonally and with the 11-year solar cycle.

▼ ? Define MUF, LUF and OWF.

Model Answer:

MUF — the maximum usable frequency the ionosphere will refract back; above it the wave is lost. LUF — the lowest usable frequency; below it the signal is absorbed. OWF — the optimum working frequency, about 85% of the MUF.

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▼ ? Why does the choice of HF frequency change between day and night?

Model Answer:

By day the D layer absorbs lower frequencies, so a higher frequency is needed; by night the D layer disappears, so a lower frequency works best.

▼ ? What is dead space, and how does it relate to skip distance?

Model Answer:

Dead space (skip zone) is the region between the limit of ground-wave reception and the point where the first sky wave returns (the skip distance) — where neither wave is received.

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▼ ? How is VHF line-of-sight range estimated, and what increases it?

Model Answer:

By the radio-horizon formula $d \approx 1.23 \times \sqrt{h}$, with h in feet; combine the heights of aircraft and station. Range increases with height — the higher you fly, the farther the horizon.

Part C — Numerical problems (tap for worked solutions)

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▼ P1. Find the radio horizon from 36,000 ft.

Solution:

d ≈ 1.23 × √36,000 = 1.23 × 189.7 ≈ **233 NM**.

▼ P2. An aircraft at 25,000 ft talks to a station antenna at 100 ft. Estimate the maximum communication range.

Solution:

aircraft 1.23 × √25,000 = 1.23 × 158.1 ≈ 194 NM; station 1.23 × √100 = 12.3 NM; total ≈ **206 NM**.

▼ P3. If the MUF is 12 MHz, estimate the OWF.

Solution:

OWF ≈ 0.85 × 12 = **10.2 MHz**.

⚡ 60-SECOND REVISION CARD

Ground (LF/MF, NDB, better over sea) · **Sky** (HF, ionosphere, multi-hop) · **Space** (VHF+, line-of-sight).

Layers **D E F1 F2**; D absorbs by day. Ionisation varies day/season/solar cycle.

HF: **day high, night low**. **MUF** above = lost · **LUF** below = absorbed · **OWF ≈ 85% MUF**.

Dead space = gap before first sky wave; skip distance grows with frequency/at night.

Radio horizon **d ≈ 1.23√h(ft)**; **ground shadow** blocks VHF behind terrain; **fading** = multi-path.

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